

EL MAHDI KHTRIBCH

PhD Candidate in Statistics • ESSEC Business School

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PROFILE

Final-year PhD candidate at ESSEC Business School, supervised by Pr. Pierre Alquier and Pr. Pierre E. Jacob. My research interests lie in *computational statistics* and *Bayesian inference*, with a focus on unbiased MCMC, couplings of Markov chains, variational inference, and PAC-Bayesian theory.

EDUCATION

Ph.D. in Statistics and Data Analytics	ESSEC Business School, Paris	2022 – present
<i>Supervised by Pr. Pierre Alquier and Pr. Pierre E. Jacob.</i>		
M.Sc. Mathematics, Vision & Learning (MVA)	ENS Paris-Saclay, Paris	2021
M.Sc. Statistical Sciences	University of Oxford, UK	2020
Visiting Year	Harvard University, Cambridge, USA	2019
M.Sc. & Executive Engineering	Mines ParisTech, Paris	2018

PUBLICATIONS

Robust Bayesian Inference via Variational Approximations of Generalized ρ-Posteriors	
El Mahdi Khribch, Pierre Alquier	
<i>Submitted</i>	arXiv:2601.07325
On Importance Sampling and Independent Metropolis–Hastings with an Unbounded Weight Function	
George Deligiannidis, Pierre E. Jacob, El Mahdi Khribch, Guanyang Wang	
<i>Major revision at The Annals of Statistics</i>	arXiv:2411.09514
Convergence of Statistical Estimators via Mutual Information Bounds	
El Mahdi Khribch, Pierre Alquier	
<i>Submitted</i>	arXiv:2412.18539

AWARDS & HONORS

Best Poster Award	Bayes Comp 2025, Singapore	2025
Major-Excellence Scholarship	French Government – 37,830 €, top 40 students	2018
Top 1% Nationwide	Concours National Commun, Morocco	2018

RESEARCH EXPERIENCE

Visiting PhD Student	Department of Statistics, University of Warwick	Feb – Mar 2026
M.Sc. Dissertation	<i>On PAC-Bayesian bounds: faster rates in hostile grounds</i> – ENS Paris-Saclay	2021
M.Sc. Dissertation	<i>On the mixing times of the Metropolized Algorithm with Optimization step (MAO): a new framework</i> – University of Oxford	2020
Research Assistant	Department of Statistics, Harvard University	2019

TEACHING EXPERIENCE

Teaching Assistant – ENPC	High-Dimensional Statistics	2023 – 2026
Teaching Assistant – ENSAE	Simulation and Monte Carlo Methods	2023 – 2024
Teaching Assistant – ENSAE	Introduction to Stochastic Processes	2023 – 2024

SKILLS

Programming	Python, R, $\text{\LaTeX}$, TensorFlow, PyTorch.
Languages	English (bilingual), French (native), Arabic (native), Spanish (B1).

ADDITIONAL INFORMATION

Sports	Football, basketball (regional college league).
Arts & Culture	Cinema, theatre, poetry; member of the PSL University debating club.